

BF27.6

1. By Lenz's Law, by decreasing the magnetic flux through the loop, the magnet induces a current in the wire that opposes the change in magnetic flux. The current flows into the page at the top of the ring and out of the page at the bottom. This means the ring repels the magnet.
2. Knowns: $m = 0.002 \text{ kg}$, $I = 1.5 \text{ A}$, $d = 0.12 \text{ m}$
Unknowns: $B = ?$

$$\begin{aligned} |F_g| &= |F_b| \\ mg &= IdB \\ B &= \frac{mg}{Id} \\ &= \frac{(0.002 \text{ kg})(10 \text{ N/kg})}{(1.5 \text{ A})(0.12 \text{ m})} \\ &= 0.11 \text{ T} \end{aligned}$$

The magnetic field points out of the page by the right hand rule.

3. Knowns: diagram, $\rho = 0.051 \text{ kg/m}$, $d = 0.04 \text{ m}$
Unknowns: $I = ?$

$$\begin{aligned} F_g &= F_b \\ mg &= IlB \\ \rho g &= IB \\ B &= \frac{\mu_0}{2\pi} \frac{I}{r} \\ &= (2 \cdot 10^{-7} \text{ H/m}) \frac{I}{0.04 \text{ m}} \left(\frac{4}{\sqrt{3}} \right) \\ &= (1.15 \cdot 10^{-5} \text{ H/m}^2) I \\ \rho g &= (1.15 \cdot 10^{-5} \text{ H/m}^2) I^2 \\ I &= \boxed{240 \text{ A}} \end{aligned}$$

4. Knowns: $I = 550 \text{ mA}$, $s = 0.05 \text{ m}$, $B = 1.3 \text{ T}$, $\theta = 30^\circ$
Unknowns: $\tau = ?$

$$\begin{aligned} |F| &= IlB \\ &= (0.55 \text{ A})(0.05 \text{ m})(1.3 \text{ T}) \\ &= 0.03575 \text{ N} \\ \tau &= rF \sin(\theta) \\ \tau &= 2(0.025 \text{ m})(0.03575 \text{ N}) \sin(30^\circ) \\ &= \boxed{0.00089 \text{ N}\cdot\text{m}} \end{aligned}$$

5. Knowns: $I_1 = 2 \text{ A}$, $I_2 = 0.2 \text{ A}$, $r = 0.001 \text{ m}$, $d = 0.02 \text{ m}$

Unknowns: $\tau = ?$, *equilibrium* = ?

$$\begin{aligned}
 B_1 &= \frac{\mu_0}{2\pi} \frac{I}{r} \\
 &= 2 \cdot 10^{-7} \text{ H/m} \frac{(2 \text{ A})}{(0.02 \text{ m})} \\
 &= 0.00002 \text{ T} \\
 \tau &= IAB \sin(\theta) \\
 &= (0.2 \text{ A})(\pi(0.01 \text{ m})^2)(0.00002 \text{ T}) \sin(90^\circ) \\
 &= \boxed{1.26 \cdot 10^{-11} \text{ N}\cdot\text{m}}
 \end{aligned}$$

The wire's equilibrium orientation is horizontal to the wire because it is in the plane of all of the magnetic field created by the other wire, making it stable.

6. Knowns: $I = 4.4 \text{ A}$, $r = 0.01 \text{ m}$

Unknowns: $B = ?$, *direction* = ?

$$\begin{aligned}
 dB &= \frac{\mu_0}{4\pi} \frac{I(dl \times r)}{r^2} \\
 B &= (1 \cdot 10^{-7} \text{ H/m}) \int_C \frac{I}{r^2} dl \\
 &= (1 \cdot 10^{-7} \text{ H/m}) \frac{(4.4 \text{ A})(2\pi(0.01 \text{ m}))}{(0.01 \text{ m})^2} \\
 &= \boxed{3.6 \cdot 10^{-4} \text{ T}}
 \end{aligned}$$

By the right hand rule, the current points into the page.

7. Knowns: $B = 0.1 \text{ T}$, $l = 0.08 \text{ m}$, $r = 0.01 \text{ m}$, $I = 1.6 \text{ A}$

Unknowns: $N = ?$

$$\begin{aligned}
 B_{sol} &= \mu_0 n I = \mu_0 \frac{N}{l} I \\
 N &= \frac{B_{sol} * l}{\mu_0 * I} \\
 &= \frac{(0.1 \text{ T})(0.08 \text{ m})}{(4 \pi \cdot 10^{-7} \text{ H/m})(1.6 \text{ A})} \\
 &= \boxed{4000}
 \end{aligned}$$

8. Knowns: $\mu = 8 \cdot 10^{22} \text{ A}\cdot\text{m}^2$, $r_1 = 1500 \text{ km}$, $r_2 = 500 \text{ km}$

Unknowns: $I = ?$, $J = ?$, $\frac{J_{loop}}{J_{wire}} = ?$

$$\begin{aligned}
 \mu &= IA \\
 8 \cdot 10^{22} \text{ A} \cdot \text{m}^2 &= I(\pi(1500000 \text{ m})^2) \\
 I &= \boxed{1.1 \cdot 10^{10} \text{ A}} \\
 J &= \frac{I}{A} \\
 &= \frac{1.1 \cdot 10^{10} \text{ A}}{\pi(500000 \text{ m})^2} \\
 &= \boxed{0.01401 \text{ A/m}^2} \\
 \frac{J_{loop}}{J_{wire}} &= \frac{0.014 \text{ A/m}^2}{2 \text{ A}/(\pi(0.0006 \text{ m})^2)} \\
 &= \boxed{8.1 \cdot 10^{-9}}
 \end{aligned}$$

9. Knowns: $v = 0.7 \cdot 10^7 \text{ m/s}$, $d = 0.01 \text{ m}$, $V = 200 \text{ V}$, $q = -1.6 \cdot 10^{-19} \text{ C}$
 Unknowns: $B = ?$, $direction_B = ?$

$$\begin{aligned}
 F_e &= F_b \\
 qE &= qvB \\
 E &= vB \\
 \frac{V}{d} &= vB \\
 B &= \frac{V}{dv} \\
 &= \frac{200 \text{ V}}{(0.01 \text{ m})(0.7 \cdot 10^7 \text{ m/s})} \\
 &= \boxed{0.00286 \text{ T}}
 \end{aligned}$$

Since the electric field accelerates the electron towards the positive plate which is up, the magnetic field must accelerate the electron towards the negative plate, or down. By the opposite of the left hand rule, the magnetic field must point into the page.